

Signed numbers: addition and subtraction

Initial assessment — answers

1. $-2 + 9$

Answer: 7

$$-2 + 2 + 7 = 7.$$

2. $6 - 11$

Answer: -5

$$6 - 6 - 5 = -5.$$

3. $-7 - (-10)$

Answer: 3

$$-7 + 10 = 3.$$

4. $5 + (-9) - (-6)$

Answer: 2

$$5 - 9 + 6 = -4 + 6 = 2.$$

Signed numbers: addition and subtraction

Answers • Step-by-step practice 1

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
$-6 + 9$ Answer: 3 $-6 + 6 + 3 = 3$; check $3 - 9 = -6$.	$4 + (-7)$ Answer: -3 $4 + (-7) = 4 - 7 = -3$; check $-3 + 7 = 4$.	$-8 + 5 - (-4)$ Answer: 1 $-8 + 5 + 4 = -3 + 4 = 1$; check $1 - 4 - 5 = -8$.

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method $-6 + 6 + 3 = 3$; check $3 - 9 = -6$.	Method $4 + (-7) = 4 - 7 = -3$; check $-3 + 7 = 4$.	Method $-8 + 5 + 4 = -3 + 4 = 1$; check $1 - 4 - 5 = -8$.
If you got... -15 : check whether you used $-6 - 9 = -15$. Instead, adding positive 9 means moving right.	If you got... 11 : check whether you used $4 + 7 = 11$. Instead, the added number is negative.	If you got... -7 : check whether you used $-8 + 5 - 4 = -7$. Instead, subtracting -4 means adding 4.
Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.	Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.	Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.

Signed numbers: addition and subtraction

Answers • Step-by-step practice 2

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
3 − 8 Answer: −5 3 + (−8) = −5; check −5 + 8 = 3.	−5 − (−9) Answer: 4 −5 + 9 = 4; check 4 + (−9) = −5.	7 − (−2) + (−12) Answer: −3 7 + 2 − 12 = 9 − 12 = −3; check −3 + 12 − 2 = 7.

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method $3 + (-8) = -5$; check $-5 + 8 = 3$.	Method $-5 + 9 = 4$; check $4 + (-9) = -5$.	Method $7 + 2 - 12 = 9 - 12 = -3$; check $-3 + 12 - 2 = 7$.
If you got... 5: check whether you used $8 - 3 = 5$. Instead, keep the original order: $3 - 8$.	If you got... -14: check whether you used $-5 - 9 = -14$. Instead, subtracting -9 means adding 9.	If you got... -7: check whether you used $7 - 2 - 12 = -7$. Instead, subtracting -2 means adding 2.
Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.	Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.	Check Compare your steps with the method above. Correct the first wrong step and redo the calculation.

Signed numbers: addition and subtraction

Answers • Independent practice 1

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 5	1. 7	1. 6
2. -5	2. -2	2. -1
3. -2	3. -10	3. 2
4. -7	4. -5	4. -3
5. 0	5. 0	5. -2
6. -6	6. -7	6. 0

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q3: $-6 + 4 = -2$; move four places right.	Method Q3: $-4 + (-6) = -10$; move six places left.	Method Q6: $3 - 8 + 5 = -5 + 5 = 0$.
If you got... Q3: 2 — you may have calculated $6 - 4 = 2$. The result stays negative because four steps do not reach zero.	If you got... Q3: 10 — you may have calculated $4 + 6 = 10$. Both numbers are negative; adding -6 moves further left.	If you got... Q6: -10 — you may have calculated $3 - 8 - 5 = -10$. Subtracting -5 means adding 5.
Check Q3: $-2 - 4 = -6$. Correct the first wrong step and redo the question.	Check Q3: $-10 + 6 = -4$. Correct the first wrong step and redo the question.	Check Q6: $0 - 5 = -5$; $-5 + 8 = 3$. Correct the first wrong step and redo the question.

Signed numbers: addition and subtraction

Answers • Independent practice 2

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. -10	1. 13	1. -1
2. 0	2. -13	2. 0
3. -9	3. 9	3. 2
4. 3	4. -5	4. -7
5. -5	5. 5	5. 3
6. -7	6. -4	6. -4

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q2: $-11 + 11 = 0$; the numbers are opposites.	Method Q3: $0 - (-9) = 0 + 9 = 9$.	Method Q1: $-6 + 9 - 4 = 3 - 4 = -1$.
If you got... Q2: -22 — you may have calculated $-11 - 11 = -22$. Adding positive 11 moves right to zero.	If you got... Q3: -9 — you may have calculated $0 - 9 = -9$. Subtracting -9 moves right, even when starting at zero.	If you got... Q1: 7 — you may have calculated $-6 + 9 + 4 = 7$. The final number being added is -4 , not $+4$.
Check Q2: $0 - 11 = -11$. Correct the first wrong step and redo the question.	Check Q3: $9 + (-9) = 0$. Correct the first wrong step and redo the question.	Check Q1: $-1 + 4 = 3$; $3 - 9 = -6$. Correct the first wrong step and redo the question.

Signed numbers: addition and subtraction

Answers • Independent practice 3

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 4	1. -11	1. -6
2. -9	2. 13	2. 0
3. -4	3. -3	3. 11
4. -5	4. -5	4. -5
5. -9	5. 6	5. 5
6. 0	6. -5	6. -5

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q2: $-5 - 4 = -9$; move four places left.	Method Q5: $-4 - (-10) = -4 + 10 = 6$.	Method Q2: $6 - 9 + 3 = -3 + 3 = 0$.
If you got... Q2: -1 — you may have calculated $-5 + 4 = -1$. Subtracting positive 4 moves left.	If you got... Q5: -14 — you may have calculated $-4 - 10 = -14$. Subtracting -10 means adding 10; this crosses zero.	If you got... Q2: -6 — you may have calculated $6 - 9 - 3 = -6$. Keep the intermediate -3 ; subtracting -3 then adds 3.
Check Q2: $-9 + 4 = -5$. Correct the first wrong step and redo the question.	Check Q5: $6 + (-10) = -4$. Correct the first wrong step and redo the question.	Check Q2: $0 - 3 = -3$; $-3 + 9 = 6$. Correct the first wrong step and redo the question.

Signed numbers: addition and subtraction

Answers • Independent practice 4

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. -5	1. 6	1. 3
2. -7	2. -3	2. -3
3. -14	3. -8	3. 0
4. 5	4. 5	4. -6
5. 0	5. 15	5. -7
6. -5	6. -13	6. 3

Signed numbers: addition and subtraction

Check your method • Correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q6: $8 - 13 = -5$; keep the subtraction in its original order.	Method Q6: $-4 + (-9) = -13$.	Method Q4: $5 - 13 + 2 = -8 + 2 = -6$.
If you got... Q6: 5 — you may have calculated $13 - 8 = 5$. Do not swap the numbers to avoid a negative result.	If you got... Q6: -5 — you may have calculated $4 - 9 = -5$. Keep the initial negative sign; both added numbers are negative.	If you got... Q4: -10 — you may have calculated $5 - 13 - 2 = -10$. Subtracting -2 means adding 2.
Check Q6: $-5 + 13 = 8$. Correct the first wrong step and redo the question.	Check Q6: $-13 + 9 = -4$. Correct the first wrong step and redo the question.	Check Q4: $-6 - 2 = -8$; $-8 + 13 = 5$. Correct the first wrong step and redo the question.